

CPSC 304 Project Cover Page

Milestone #: 4

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Group Number: 26

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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above.

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia.

Our project is a web application for tracking machine learning experiments. The database stores projects and their development phases, experimental runs linked to datasets, hardware configurations, models, hyperparameters, and performance metrics. The application was built using Node.js and Express on the backend with the department provided Oracle database, and React with TypeScript on the frontend. It provides a clean UI for managing and querying experiment data. The final schema does not differ from the original schema we turned in.

The visual representation of the state of the database is located at the bottom of the document.

All SQL queries are implemented in appService.js. Each function contains the SQL statement executed against the Oracle database via the oracledb connection pool.

Non-hardcoded queries:

1. INSERT: lines 150-170
2. UPDATE: lines 317-319
3. DELETE: line 346
4. SELECT: lines 429-471
5. PROJECT: line 539
6. JOIN: lines 492-497

Hardcoded queries:

7. Aggregation with GROUP BY: lines 558-561
 - Returns the total number of runs for each project by grouping all runs on their project ID.
 - ```
SELECT project_id, COUNT(*) AS run_count
FROM Run
GROUP BY project_id
ORDER BY project_id
```
8. Aggregation with HAVING: lines 583-587
  - Returns only projects whose total run count meets or exceeds a specified minimum.
  - ```
SELECT project_id, COUNT(*) AS run_count
FROM Run
GROUP BY project_id
HAVING COUNT(*) >= :minRuns
ORDER BY project_id
```

9. Nested Aggregation with GROUP BY: lines 608-620

- Returns the project(s) with the highest average metric value for a selected metric type.

- ```
SELECT r.project_id, AVG(m.metric_value) AS avg_metric
```

```
FROM Run r
```

```
JOIN Metric m ON r.run_id = m.run_id
```

```
WHERE m.metric_name = :metric_name
```

```
GROUP BY r.project_id
```

```
HAVING AVG(m.metric_value) >= ALL (
```

```
 SELECT AVG(m2.metric_value)
```

```
 FROM Run r2
```

```
 JOIN Metric m2 ON r2.run_id = m2.run_id
```

```
 WHERE m2.metric_name = :metric_name
```

```
 GROUP BY r2.project_id
```

```
)
```

```
ORDER BY r.project_id
```

10. Division: lines 637-651

- Returns every run that has a usage record for all hyperparameters marked as required.

- ```
SELECT r.run_id
```

```
FROM Run r
```

```
WHERE NOT EXISTS (
```

```
    SELECT h.parameter_id
```

```
    FROM Hyperparameter h
```

```
    WHERE h.is_required = 'Y'
```

```
    AND NOT EXISTS (
```

```
        SELECT 1
```

```
        FROM Uses u
```

```
        WHERE u.run_id = r.run_id
```

```
        AND u.parameter_id = h.parameter_id
```

```
)
```

```
)
```

```
ORDER BY r.run_id
```

Database State

Table: Author

email	organization	role	fullname
mark@email.com	UBC	Student	Mark John
adi@email.com	UBC	Teaching Assistant	Adi Yadav
chaitanya@email.com	UBC	Student	Chaitanya John
john@email.com	SFU	Professor	John Joe
frank@email.com	UVIC	Student	Frank Johnson
bob@email.com	UBC	Research Assistant	Bob Dylan

Table: Project

project_id	name	deadline	start_date
1	House Price Regression	2026-04-20	2026-01-05
2	Customer Churn Classification	2026-04-22	2026-01-08
3	Image Clustering Study	2026-04-25	2026-01-12
4	Energy Forecasting	2026-04-27	2026-01-15
5	Fraud Detection	2026-04-29	2026-01-18
6	Document Topic Modeling	2026-05-02	2026-01-20

Table: PhaseComplexity

phase_name	complexity_score
Data Collection	2
Preprocessing	3
Training	5
Evaluation	3
Deployment	4

Table: Phase

phase_name	project_id	estimated_hours	description
Data Collection	1	10	Collect housing and sales data
Preprocessing	1	14	Clean missing values and scale features
Training	1	20	Train regression models
Evaluation	1	8	Evaluate MAE and R2
Data Collection	2	8	Gather customer churn records
Preprocessing	2	12	Balance classes and normalize data
Training	2	18	Train classification models
Evaluation	2	9	Compare accuracy and F1
Data Collection	3	7	Collect image embeddings
Preprocessing	3	10	Normalize clustering features
Training	3	16	Run clustering algorithms
Evaluation	3	7	Check silhouette scores
Data Collection	4	9	Collect energy consumption history
Preprocessing	4	13	Build forecasting features
Training	4	17	Train forecasting models
Evaluation	4	8	Evaluate MAE and RMSE
Data Collection	5	11	Collect fraud transaction logs
Preprocessing	5	15	Remove anomalies and encode features
Training	5	21	Train fraud detection classifiers
Evaluation	5	10	Evaluate precision and F1
Data Collection	6	6	Collect text corpora
Preprocessing	6	9	Convert documents to embeddings
Training	6	14	Run topic clustering
Evaluation	6	6	Evaluate topic coherence

Table: Hyperparameter

parameter_id	hyperparam_name	default_value	is_required	datatype
1	learning_rate	0.01	Y	FLOAT
2	batch_size	32	Y	INTEGER
3	epochs	10	Y	INTEGER
4	dropout	0.2	N	FLOAT
5	max_depth	5	N	INTEGER
6	decision_threshold	0.5	N	FLOAT
7	num_clusters	8	N	INTEGER

Table: Dataset

dataset_id	format	row_count	feature_count	creation_date
1	CSV	12000	24	2026-01-01
2	CSV	18000	31	2026-01-03
3	PARQUET	26000	42	2026-01-05
4	IMAGES	9000	128	2026-01-07
5	CSV	21000	18	2026-01-09
6	JSON	15000	36	2026-01-11

Table: HardwareConfiguration

config_id	gpu_type	cpu_cores	memory_GB
1	RX 7900	8	16
2	NVIDIA H100	16	32
3	NVIDIA A100	32	64
4	RTX 4090	24	128
5	RTX 5090	32	256

Table: Model

model_id	file_path	format	size_bytes
1	/models/reg_house_v1	MODEL	1000
2	/models/reg_house_v2	MODEL	1100
3	/models/forecast_v1	MODEL	1200
4	/models/churn_cls_v1	MODEL	1300
5	/models/churn_cls_v2	MODEL	1400
6	/models/fraud_cls_v1	MODEL	1500
7	/models/image_clust_v1	MODEL	1600
8	/models/image_clust_v2	MODEL	1700
9	/models/topic_clust_v1	MODEL	1800

Table: Regression

model_id	target_transform
1	log
2	none
3	standardize

Table: Classification

model_id	decision_threshold
4	0.5
5	0.55
6	0.6

Table: Clustering

model_id	algorithm_family
7	K-Means
8	DBSCAN
9	Hierarchical

Table: Creates

email	project_id
mark@email.com	1
adi@email.com	1
chaitanya@email.com	2
adi@email.com	2
john@email.com	3
bob@email.com	3
adi@email.com	4
frank@email.com	4
john@email.com	5
chaitanya@email.com	5
bob@email.com	6

Table: Run

run_id	start_time	end_time	execution_status	project_id	model_id	dataset_id	config_id
1	2026-01-15 09:00:00	2026-01-15 09:45:00	COMPLETED	1	1	1	1
2	2026-01-16 10:00:00	2026-01-16 10:50:00	COMPLETED	1	2	2	2
3	2026-01-17 11:00:00	2026-01-17 11:40:00	COMPLETED	2	4	2	2
4	2026-01-17 12:00:00	2026-01-17 15:40:00	COMPLETED	2	4	2	2
5	2026-01-26 12:00:00	2026-01-26 12:47:00	COMPLETED	2	2	2	2
6	2026-01-18 11:30:00	2026-01-18 12:05:00	FAILED	2	5	3	3
7	2026-01-19 13:00:00	2026-01-19 13:48:00	COMPLETED	3	7	4	1
8	2026-01-20 14:10:00		RUNNING	3	8	4	4
9	2026-01-21 09:15:00	2026-01-21 10:05:00	COMPLETED	4	3	5	3
10	2026-01-22 15:00:00	2026-01-22 15:42:00	COMPLETED	5	6	3	2
11	2026-01-23 16:05:00	2026-01-23 16:56:00	COMPLETED	5	4	6	4
12	2026-01-24 08:35:00	2026-01-24 09:20:00	FAILED	6	9	6	1
13	2026-01-25 10:20:00	2026-01-25 11:05:00	COMPLETED	4	1	5	5
14	2026-01-24 08:35:00	2026-01-26 12:47:00	COMPLETED	2	3	3	2

Table: Uses

run_id	parameter_id	hyperparam_value
1	1	0.010
1	2	32
1	3	12
1	4	0.15
2	1	0.008
2	2	64
2	3	16
2	4	0.20
2	5	8
3	1	0.005
3	2	128
3	3	10
3	6	0.50
4	1	0.003
4	2	64
4	6	0.55
5	2	32
5	3	20
5	7	6
6	1	0.020
6	2	32
6	3	15
6	7	8
7	1	0.006

7	3	18
7	5	10
8	1	0.002
8	2	256
8	3	8
8	6	0.62
9	1	0.004
9	2	128
9	3	14
9	4	0.10
9	6	0.58
10	2	64
10	3	9
10	7	12
11	1	0.007
11	2	32
11	3	14
11	5	7
12	1	0.009
12	2	48
12	3	11
12	7	10

Table: MetricType

metric_name	unit	goal	min_value	max_value
mae	score	MIN	0	100
mse	score	MIN	0	1000
rmse	score	MIN	0	100
r2	score	MAX	0	1
loss	score	MIN	0	100
accuracy	score	MAX	0	1
f1	score	MAX	0	1
silhouette	score	MAX	-1	1

Table: Metric

metric_id	metric_name	run_id	metric_value
1	mae	1	8.9
2	mse	1	70
3	rmse	1	8.4
4	r2	1	0.81
5	loss	1	0.35
6	mae	2	7.8
7	mse	2	62
8	rmse	2	7.9
9	r2	2	0.84
10	loss	2	0.28
11	accuracy	3	0.89
12	f1	3	0.86
13	loss	3	0.42
14	accuracy	4	0.76
15	f1	4	0.72
16	loss	4	0.95
17	silhouette	5	0.61
18	loss	5	0.33
19	accuracy	6	0.91
20	f1	6	0.9
21	loss	6	0.27
22	mae	7	6.8
23	mse	7	55

24	rmse	7	7.4
25	r2	7	0.88
26	loss	7	0.22
27	accuracy	8	0.87
28	f1	8	0.84
29	loss	8	0.31
30	accuracy	9	0.93
31	f1	9	0.91
32	loss	9	0.24
33	silhouette	10	0.58
34	loss	10	0.49
35	mae	11	6.3
36	mse	11	49
37	rmse	11	7
38	r2	11	0.9
39	loss	11	0.2
40	silhouette	12	0.66
41	loss	12	0.29